CleanLeaf

Eat Clean & Earn Green

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Whitman Dean's Sustainability Challenge (2026)

Executive Summary

Across corporations and universities worldwide, wellness programs consistently fail to drive meaningful participation. Employees and students are asked to download apps, track habits on separate platforms, and engage with tools that exist outside their daily routines. The result is poor nutrition and lifestyle habits and billions of dollars in productivity losses attributable to that[1].

CleanLeaf is a community wellness management platform for universities and workplaces that eliminates this friction entirely. Institutions pay for the platform; their people benefit from it. Using Generative AI and verified nutritional databases, CleanLeaf builds personalized wellness challenges tailored to each community. Participants receive a simple message on WhatsApp, Teams, or iMessage, tap a URL, photograph their meal, and an object-recognition AI model logs it automatically. No app download. No sign-up. No behavior change required to engage.

The admin dashboard gives institutional clients full CRM-style control importing contacts, segmenting cohorts, scheduling targeted messages, and tracking outcomes in real time. Every data point maps directly to the organization's Sustainable Development Goals, producing board-ready ESG reports that justify the investment.

CleanLeaf operates on a tiered SaaS pricing model, scaling from student organizations to enterprise clients. The platform targets a $61.9 billion global corporate wellness market growing at 5.9% annually[2], with an immediate beachhead in universities and mid-size corporations seeking measurable, low-friction wellness solutions.

Problem Statement

Institutional wellness programs are failing on a massive scale. Despite widespread adoption, participation in lifestyle wellness programs averages just 7–21% of eligible employees, with half of all employers reporting participation at or below 10%.[3] A randomized controlled trial of 5,000 employees found no significant causal effect on health behaviors or medical costs after two years since programs consistently attract the already-healthy while failing those at highest risk.[4] The cost

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[1] Goetzel, R. Z., Pei, X., Tabrizi, M. J., Henke, R. M., Kowlessar, N., Nelson, C. F., & Metz, R. D. (2012). Ten modifiable health risk factors are linked to more than one-fifth of employer–employee health care spending. Health Affairs, 31(11), 2474–2484. https://doi.org/10.1377/hlthaff.2011.0819

[2] Grand View Research. (2024). Corporate wellness market size, share & trends analysis report. Grand View Research. https://www.grandviewresearch.com/industry-analysis/corporate-wellness-market

[3] Mattke, S. et al. (2013). Workplace Wellness Programs Study: Final Report. RAND Corporation, RR-254-DOL. https://www.rand.org/pubs/research_reports/RR254.html

[4] Jones, D., Molitor, D. and Reif, J. (2019). What Do Workplace Wellness Programs Do? Evidence from the Illinois Workplace Wellness Study. The Quarterly Journal of Economics, 134(4), 1747–1791. https://academic.oup.com/qje/article/134/4/1747/5550759

of inaction is severe as poor employee health costs U.S. employers $575 billion annually through absenteeism and lost productivity.[5]

Digital wellness tools have not solved the problem. A 2024 review of 525,824 users found a median 70% abandonment rate within the first 100 days of downloading a health app,[6] with open rates collapsing from 69% on day one to just 3.3% by day 30.[7] The core failure is one of design: standalone apps demand behavioral change before any habit has formed, require users to adopt a new platform and offer no social accountability. They treat wellness as an individual challenge when the evidence shows it is a community one.

The problem is especially acute on university campuses. The Healthy Minds Study 2024–2025 found that 52% of students report high levels of loneliness[8] while 28.1% of students ate less than they felt they should due to financial constraints.[9] Behavioral science confirms why interventions fail in this environment as forming a new habit takes an average of 66 days.[10] Only 24% of employees who are aware of wellness offerings actually use them because they are not embedded in the daily environments where people already spend their time.[11][12]

Solution & SDGs Addressed

CleanLeaf is a community wellness management ecosystem designed for institutions such as universities and workplaces. It empowers administrators to design, launch, and monitor dietary and lifestyle challenges for their communities, encouraging healthier and more sustainable eating habits at scale. Rather than targeting individuals in isolation, CleanLeaf operates on a community-first model where participants compete on leaderboards, and receive challenges tailored to their cohort's actual behavior patterns.

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[5] Integrated Benefits Institute (2020). Poor Health Costs US Employers $575 Billion and 1.5 Billion Days of Lost Productivity. https://news.ibiweb.org/poor-health-costs-us-employers-575-billion

[6] Kidman, P.G. et al. (2024). When and Why Adults Abandon Lifestyle Behavior and Mental Health Mobile Apps: Scoping Review. Journal of Medical Internet Research, 26, e56897. https://www.jmir.org/2024/1/e56897

[7] Baumel, A. et al. (2019). Objective User Engagement With Mental Health Apps: Systematic Search and Panel-Based Usage Analysis. Journal of Medical Internet Research, 21(9), e14567. https://www.jmir.org/2019/9/e14567/

[8] Healthy Minds Network (2025). Healthy Minds Study 2024–2025 Data Report. University of Michigan / UCLA / Boston University / Wayne State University. https://healthymindsnetwork.org/research/data-for-researchers/

[9] ACHA (2024). ACHA-NCHA IIIb Spring 2024 Reference Group Executive Summary. American College Health Association.https://www.acha.org/wp-content/uploads/NCHA-IIIb_SPRING_2024_REFERENCE_GROUP_EXECUTIVE_SUMMARY.pdf

[10] Lally, P. et al. (2010). How are habits formed: Modelling habit formation in the real world. European Journal of Social Psychology, 40(6), 998–1009. https://onlinelibrary.wiley.com/doi/10.1002/ejsp.674

[11] Matthews, G. (2015). The Impact of Commitment, Accountability, and Written Goals on Goal Achievement. 9th Annual International Conference, Athens Institute for Education and Research. https://scholar.dominican.edu/psychology-faculty-conference-presentations/3/

[12] Witters, D. and Agrawal, S. (2015). Why Your Workplace Wellness Program Isn't Working. Gallup Workplace. https://www.gallup.com/workplace/236531/why-workplace-wellness-program-isn-working.aspx

Problem: "Poor Dietary Habits in Communities" / Solution: "Social Accountability for Better Nutrition"

SDG 3[13] calls for ensuring healthy lives and promoting wellbeing for all people at all ages. One of its central targets is promoting healthy lifestyles and improving nutritional wellbeing across all communities. A large share of the global population, particularly in university and workplace settings, consumes diets high in ultra-processed foods, low in fiber and plant-based nutrients, and insufficient in hydration, all of which negatively affect physical and mental wellbeing over time. The challenge is not simply awareness; most people understand that eating well matters. The real barrier is the lack of structure, motivation, and social context that makes sustained behavior change possible. CleanLeaf addresses this directly by embedding healthy habits into the daily life of a community through challenges that are time-bound, trackable, and shared. When an entire student organization commits to a high-fiber week or a hydration goal together, the behavior becomes normalized within that group's culture, increasing the likelihood that it persists beyond the challenge itself.

Problem: "Unsustainable Food Consumption" / Solution: "Steering Communities Toward Responsible Choices"

SDG 12[14] focuses on ensuring sustainable consumption and production patterns. A significant portion of unsustainable consumption is driven by the overconsumption of animal products and ultra-processed foods. SDG 12 specifically calls for encouraging people to make informed choices about what they consume and for institutions to take responsibility for shifting those patterns at scale. CleanLeaf directly supports this goal by steering communities toward healthy and minimally processed food choices through its challenge framework. When participants log meals, avoid ultra-processed products, or commit to vegetarian eating for an extended period, they are collectively reducing demand for the most environmentally costly parts of the food system. Importantly, CleanLeaf includes built-in SDG progress tracking, which means institutions can measure and report their contribution to responsible consumption in a concrete and structured way, turning everyday dining decisions into meaningful sustainability data.

Together, these two goals are deeply interconnected. A diet that is better for human health is, in most cases, also better for the planet. CleanLeaf sits at that intersection, giving communities the structure and motivation to shift their consumption habits in ways that benefit both personal wellbeing and global sustainability.

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[13] Our World in Data team. (2023, July 18). "Ensure healthy lives and promote well-being for all at all ages." Our World in Data. https://ourworldindata.org/sdgs/good-health-wellbeing

[14] United Nations. (n.d.). "Sustainable Development Goal 12: Responsible consumption and production." Wikipedia / United Nations. https://en.wikipedia.org/wiki/Sustainable_Development_Goal_12

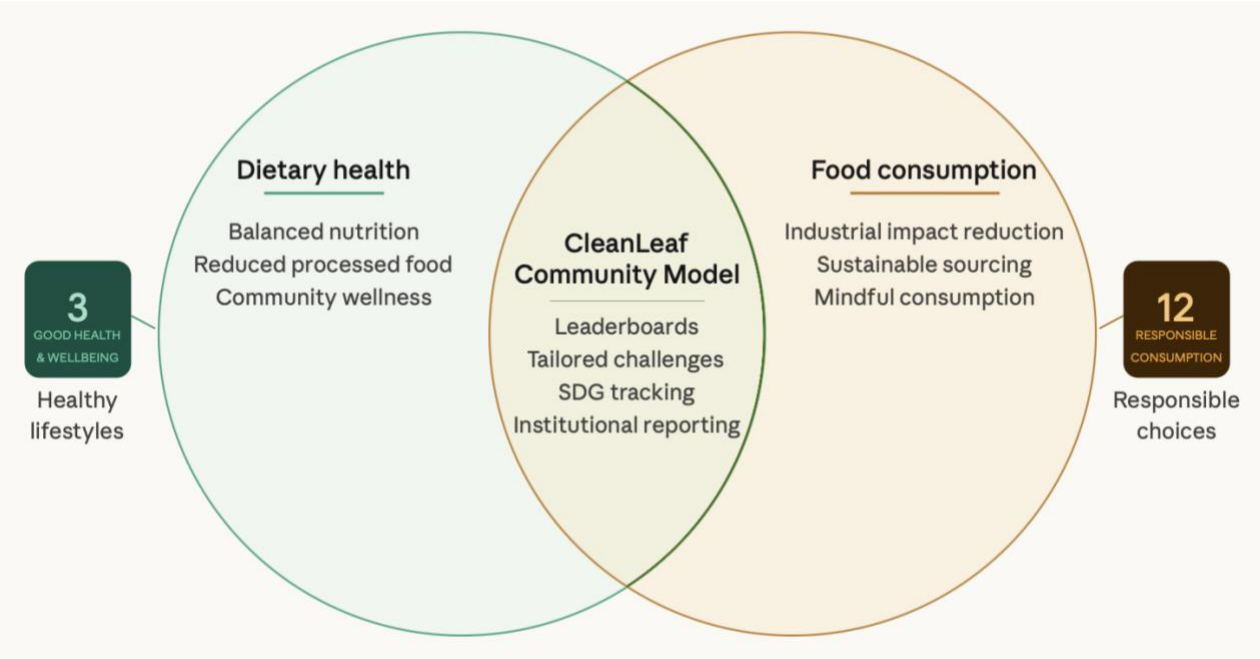

Figure 1. Venn diagram illustrating how the CleanLeaf Community Model sits at the intersection of SDG 3 (Good Health & Wellbeing) and SDG 12 (Responsible Consumption)

Prototype

Admin Dashboard — Overview

The primary interface of the CleanLeaf prototype is the admin dashboard, accessible via a secure login portal by a university administrator or corporate HR manager. Upon login, the dashboard presents a real-time overview of the organization's wellness programs across six key sections: Overview, Contacts & CRM, Challenges, Messaging, Leaderboard, and SDG Reports.

The Overview screen surfaces four headline metrics at a glance: active participants, average challenge completion rate, total messages sent, and meals logged on the current day. These are accompanied by a weekly engagement chart tracking messages, meals logged, and platform logins; a live group leaderboard; and a sustainability goal progress panel. This single-screen summary is designed to allow an administrator to assess program health without navigating away from the home-view.

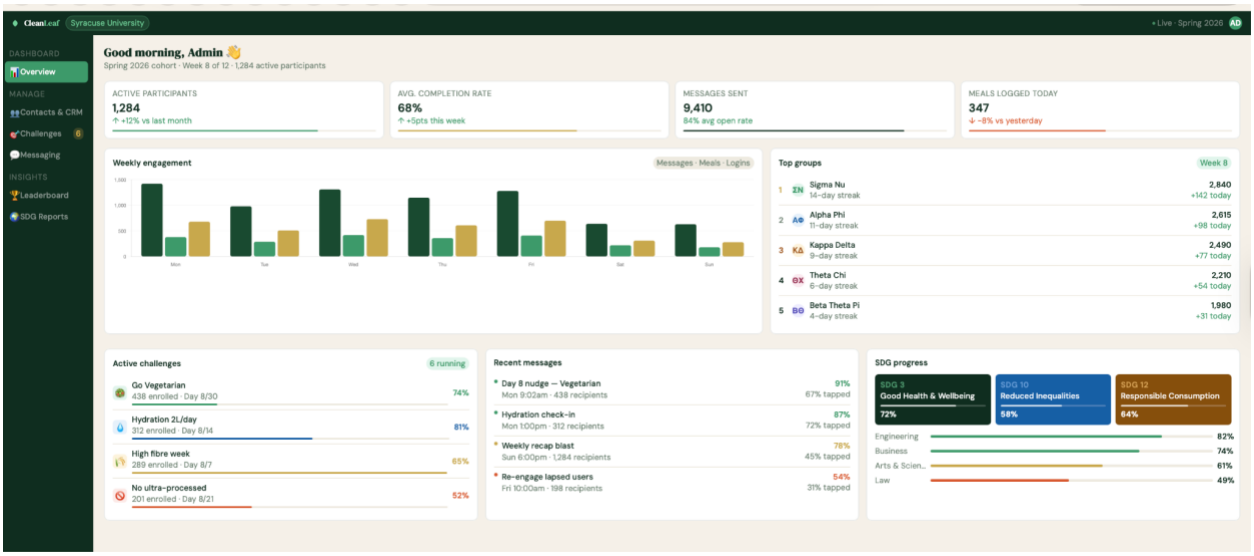

Figure 2: CleanLeaf Admin Dashboard — Overview screen showing live metrics, engagement chart, group leaderboard, and SDG progress panel.

Contacts & CRM Integration

The Contacts section demonstrates how an organization onboards its community onto CleanLeaf. Administrators can either upload a CSV or Excel file containing employee or student contact data or connect directly to an existing CRM system. The prototype showcases live integrations with four platforms:

- Salesforce
- HubSpot
- Microsoft 365
- Google Workspace

Once imported, contacts are displayed in a structured table showing each participant's name and group, active challenge, engagement status, and current streak length. This allows administrators to identify at a glance which participants are active, which are lapsing, and who may need a re-engagement message.

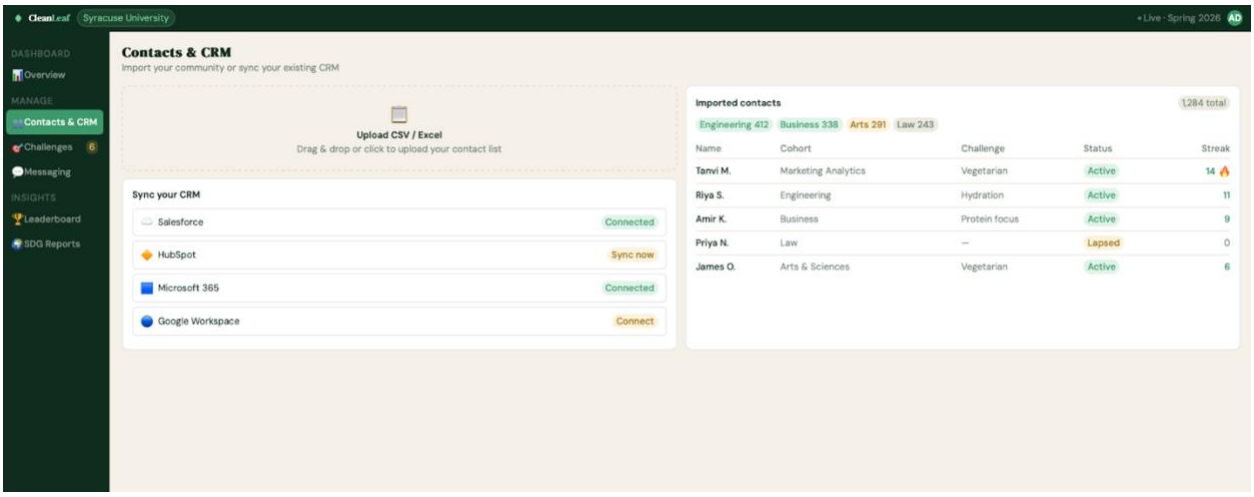

Figure 3: Contacts & CRM screen — CSV upload, CRM sync integrations, and participant contact table with streak tracking.

AI-Powered Challenge Engine

The Challenges section is the core of the CleanLeaf value proposition. The platform uses Generative AI models in combination with nutritional databases to suggest personalized wellness challenges for each community.

Beyond managing active programs, the AI engine analyses cohort-level data and surfaces new recommendations in real time.

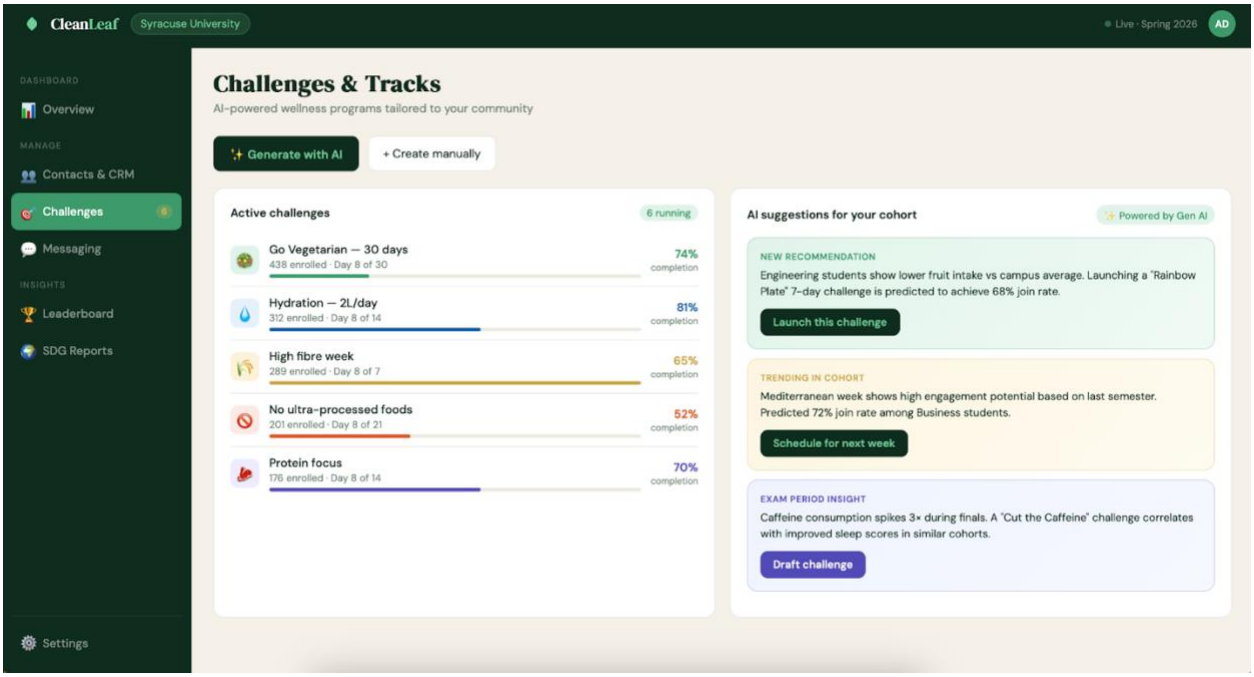

Figure 4: AI-powered Challenges screen — active programs with completion rates and live AI-generated challenge suggestions for the cohort.

Messaging & CRM-Style Targeting

The Messaging section demonstrates how CleanLeaf delivers wellness nudges to participants through their existing platforms. The compose panel allows administrators to:

- Select a target segment through a CRM-like filtering feature (e.g., all lapsed users)
- Choose a delivery channel (e.g., WhatsApp)
- Write a message that the AI personalizes per recipient based on name, challenge progress, and streak
- Schedule the message

This is the critical moment in the user journey. The participant receives a standard WhatsApp message containing a short URL in a university- or corporation-controlled group chat. They tap the link, which opens a scanner-based web application. They take a photograph of their meal, and an object-recognition AI model identifies the food item and logs the meal automatically, without app download, registration or friction.

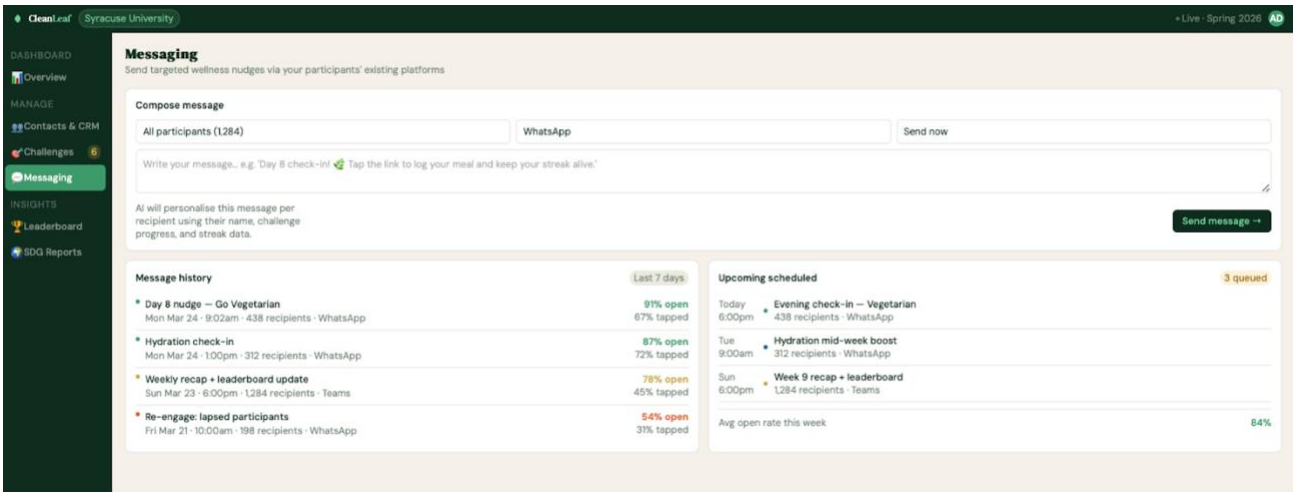

Figure 5: Messaging screen — compose panel with segment targeting and channel selection, message history with open rates and tap-through rates.

Gamification & Group Leaderboard

Social accountability is central to the CleanLeaf behavior-change model. The Leaderboard section of the prototype shows group rankings updated in real time. Groups (e.g., university sororities) compete for the highest cumulative wellness points.

These rankings are surfaced directly inside monitored group chats on the participants' messaging platforms, creating visible social comparison and peer encouragement. This mechanism transforms

an individual health activity into a shared community experience — which research consistently shows is the most effective driver of sustained habit formation.

SDG Reporting Suite

The SDG Reports section closes the loop for institutional clients. Every challenge completed, every meal logged, and every streak maintained is mapped to the organization's stated Sustainable Development Goals. launch

Cohort-level breakdowns allow administrators to identify which groups are driving impact and which require additional intervention. The report gives institutions a tangible, board-ready document demonstrating measurable ESG and sustainability outcomes — directly tied to their investment in the CleanLeaf platform.

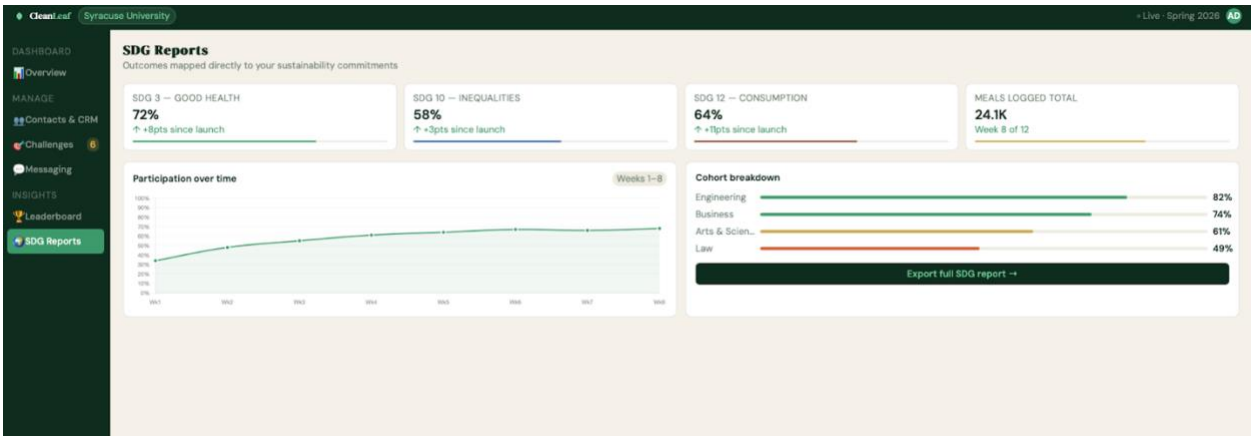

Figure 6: SDG Reports screen — SDG progress metrics, participation trend line, cohort breakdown, and one-click report export.

Prototype Summary

Component	Status in Prototype
Admin Dashboard	Fully interactive — all 6 sections clickable and functional
CRM Integration	Simulated — Salesforce, HubSpot, M365, Google Workspace
AI Challenge Engine	Simulated — live suggestions, one-click launch
Messaging System	Simulated — compose, segment, schedule, history with open rates
Object Recognition	Flow demonstrated — meal scan → identification → logging
Gamification	Functional — live leaderboard with group rankings and streaks
SDG Reporting	Functional — 3 SDG goals tracked, cohort breakdown, PDF export

Figure 7: Summary of Prototype Components

Feasibility and Limitations

Technical Feasibility

CleanLeaf's architecture is designed to minimize development complexity by building on existing infrastructure rather than creating proprietary systems from scratch. The platform's core functionality relies on four proven technology layers: a generative AI engine for challenge recommendation, open-source and commercially available image recognition models for packaged goods and meal recognition, public food composition databases for nutritional scoring, and third-party messaging APIs for user-facing delivery. Each of these components is commercially available and well-documented, which significantly reduces time-to-market risk.

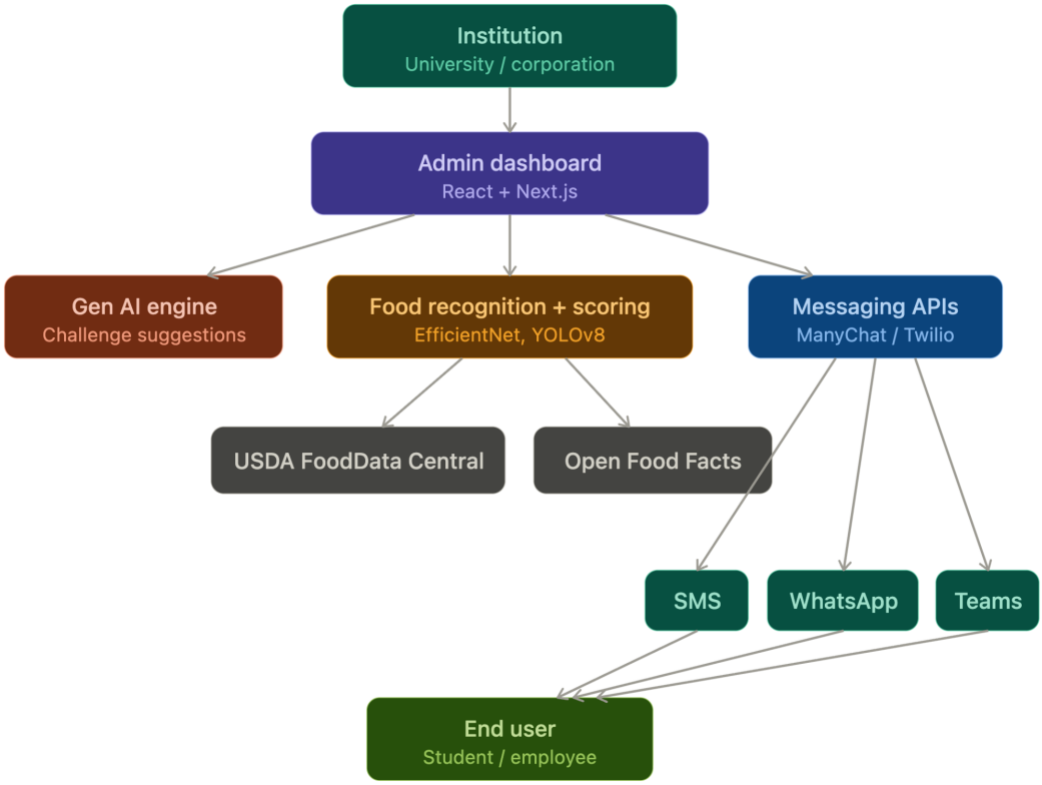

Figure 8: Flowchart of Technical Components

The admin dashboard will be a web application built using standard front-end frameworks such as React with Next.js. On the back end, meal recognition can leverage existing deep learning architectures such as EfficientNet and YOLOv8, which have been extensively applied to food image classification and detection tasks[15]. Open source food databases such as the USDA FoodData Central[16] (published under CC0 1.0 Universal) and Open Food Facts (available under

[15] Lan, N. et al. (2025). "Deep Learning in Food Image Recognition: A Comprehensive Review." *Applied Sciences*, 15(14), 7626. https://doi.org/10.3390/app15147626
[16] USDA Agricultural Research Service (n.d.). "FoodData Central API Guide." https://fdc.nal.usda.gov/api-guide/

the Open Database License with free API access[17]) provide the nutritional data needed to generate clean scores without requiring proprietary data collection.

Message distribution through platforms like WhatsApp and Microsoft Teams is achievable via established business messaging providers such as ManyChat. These providers also support automated message flows, CRM-style targeting, and delivery tracking[18].

Operational Feasibility

From an operational perspective, CleanLeaf's B2B2C model aligns incentives in a way that supports adoption. The institution (whether a university or corporation) is the paying customer and the entity responsible for launching and managing wellness initiatives. The latter are becoming increasingly relevant as ESG reporting shifts from voluntary to mandatory across major jurisdictions. Large institutions are now subject to mandatory ESG disclosure laws in the EU, UK, Canada, and parts of the United States[19]. CleanLeaf's reporting suite, which maps participation data directly to SDG targets, is specifically designed to make this value proposition tangible for administrators who must justify expenditures against these commitments.

Limitations

It is important to acknowledge the privacy concerns surrounding habit tracking and health data require careful handling. While progress is tracked privately and no personally identifiable data is shared on leaderboards, institutions operating in the European Union or handling employee health information in the United States will need to ensure compliance with GDPR and HIPAA respectively. Our platform architecture addresses this in two ways. First, our design encourages the grouping of individuals and reporting at the group level rather than the individual level. Secondly, and more importantly, we believe that CRM synchronization is the only viable path for managing data privacy at scale. Integrating with messaging providers such as WhatsApp's Business Platform requires that our backend application undergo testing and verification by the provider itself before it is approved for access to end-user data. Meta, for instance, verifies business identity, enforces platform compliance policies, and restricts or bans accounts that exhibit inconsistent data handling[20]. This provider-level verification process provides institutional clients with a reliable indicator of trust, as the providers themselves are well-known and respected entities.

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[17] Open Food Facts (n.d.). "Data, API and SDKs." https://world.openfoodfacts.org/data
[18] ManyChat (n.d.). "Connect with your Customers Instantly on WhatsApp." https://manychat.com/product/whatsapp
[19] Grace, C. (2025, September 8). "Enterprise guide to ESG reporting requirements in 2026." Pulsora. https://www.pulsora.com/blog/enterprise-guide-to-esg-reporting-requirements
[20] GMCSCO (2026, February 24). "WhatsApp Business API Compliance 2026 — Simple Guide." https://gmcsco.com/your-simple-guide-to-whatsapp-api-compliance-2026/

In addition to this foundation, CleanLeaf will pursue GDPR and HIPAA certifications to provide institutions with the formal compliance assurances their legal and procurement teams require.

Implementation Plan and Next Steps

CleanLeaf's path to market follows two phases. Phase 1 focuses on building a functional MVP and piloting it with 50–100 Syracuse University students during Fall 2026. Phase 2 refines the platform using real pilot data, completes the SDG reporting suite, and begins outreach to early-adopter institutions. Detailed phase descriptions together with relevant success metrics are provided in the Appendix.

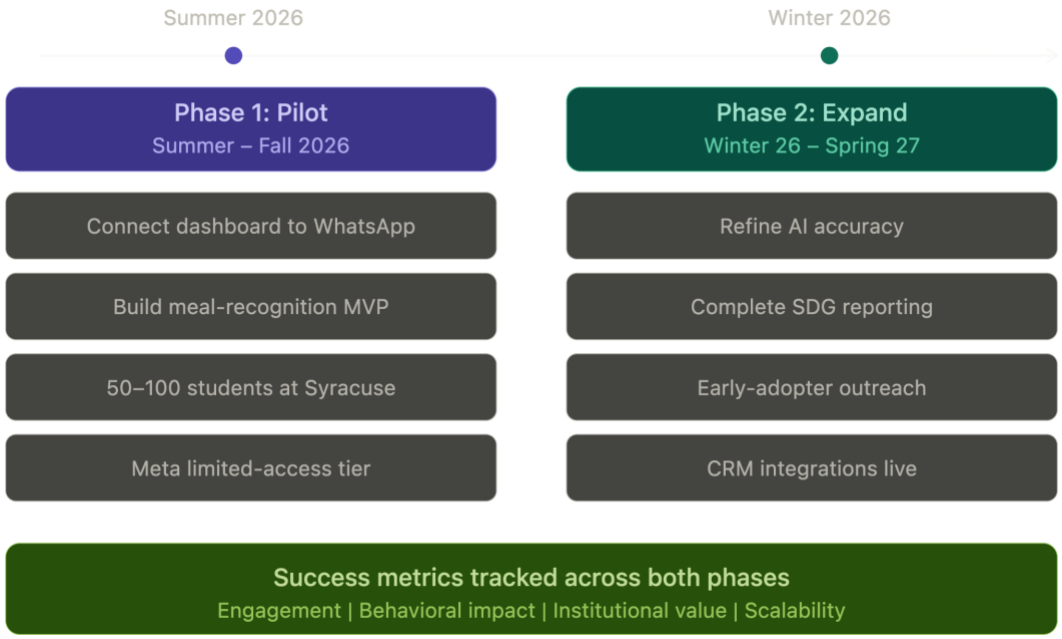

Figure 9: Next Steps for CleanLeaf

Conclusion

CleanLeaf addresses a problem that is simultaneously universal and underserved: wellness programs that exist on paper but fail in practice. Participation in corporate wellness initiatives consistently remains low not because people do not care about their health, but because the tools designed to support them create more friction than motivation. By meeting people where they already are inside the messaging platforms they use every day CleanLeaf removes the friction. Research consistently shows that habit formation is significantly stronger in social settings and that behavior change spreads most effectively through existing social networks. CleanLeaf is built precisely around this insight. The platform's B2B2C model creates a clear and sustainable

commercial structure. Institutions pay for a tool that demonstrably improves community health and generates measurable SDG-aligned outcomes they can report to stakeholders.

The prototype demonstrates that the full product is technically viable, commercially structured and immediately deployable at the institutional level. The next phase of development will focus on integrating live AI APIs, onboarding a pilot university partner and validating engagement benchmarks with real cohort data. The habit built during their formative age at an organization does not leave when they do. That lasting impact formed in their prime, sustained for life is the true measure of CleanLeaf's value.

Word Count: 2,456

AI Disclaimer: This document was edited with the assistance of AI tools, specifically Claude (Anthropic), which was used for formatting and grammar corrections. All intellectual content, research, analysis and strategic direction contained in this document were developed by the CleanLeaf team.

Appendix

Phase 1: Minimum Viable Product (Summer–Fall 2026)

The next milestone in our plan is developing a Minimum Viable Program (MVP) that can run a simple challenge with real participants by the beginning of the Fall 2026 term. The requirements for this MVP are:

- establishing a connection between the administrative dashboard and a live messaging provider, and
- developing a fundamental meal-recognition pipeline

The pilot program will be implemented at Syracuse University, enrolling 50 to 100 students from the Whitman School of Management as a convenience sample and at least one Greek-life organization. Team members will collect data through consent forms that explicitly cover data handling and communications permissions. After the pilot's conclusion, forms will be sent out to participants for feedback. The objective is to generate authentic engagement data, subject the AI scoring to rigorous pressure testing using actual meals, and identify usability issues that cannot be anticipated through design work alone.

Phase 2: Expansion (Winter 2026–Spring 2027)

Phase 2 involves addressing the issues identified in Phase 1 and refining the effective components. Key priorities include enhancing the accuracy of meal-recognition systems and streamlining

workflows using real-world data from Phase 1. Additionally, the team will be focused on completing the SDG reporting suite. At this time, the team also will begin conversations with campus wellness offices and corporate HR departments to line up early-adopter clients beyond Syracuse. The platform will integrate with major CRM systems (Salesforce, HubSpot) and enterprise communication tools (Microsoft 365, Google Workspace) to minimize friction during onboarding.

Success Metrics

Across all three phases, CleanLeaf will track four categories of metrics:

- engagement (daily active participation, message open and tap-through rates, challenge completion), behavioral impact (changes in Clean Scores over time, habit persistence after a challenge ends),
- institutional value (SDG reporting completeness, administrator satisfaction, contract renewals), and
- scalability (onboarding time per institution, system reliability under load).

These figures will inform product decisions, showcase impact to stakeholders, and establish a robust evidence base to attract investment.